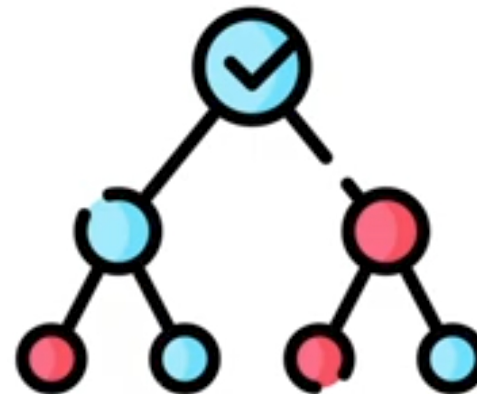


Siddhardhan

Decision Tree - intuition

the Channel



Decision Tree

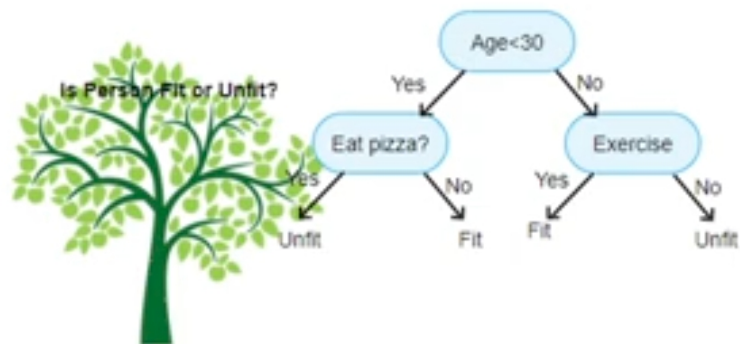
About Decision Tree model:

1. Supervised Learning Model
2. Used for both Classification & Regression
3. Builds Decision Nodes at each step
4. Basis of Tree-based models

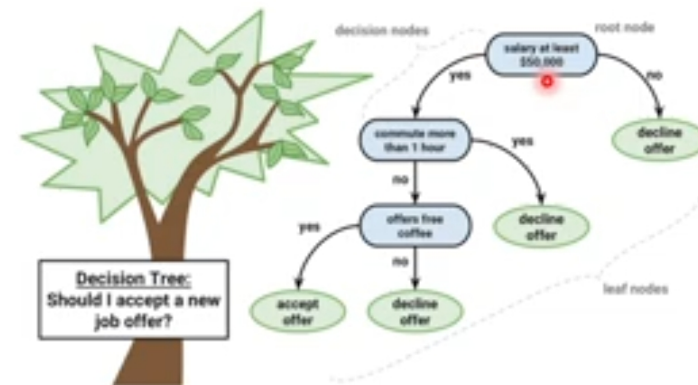


Decision Tree

Decision Tree – examples:

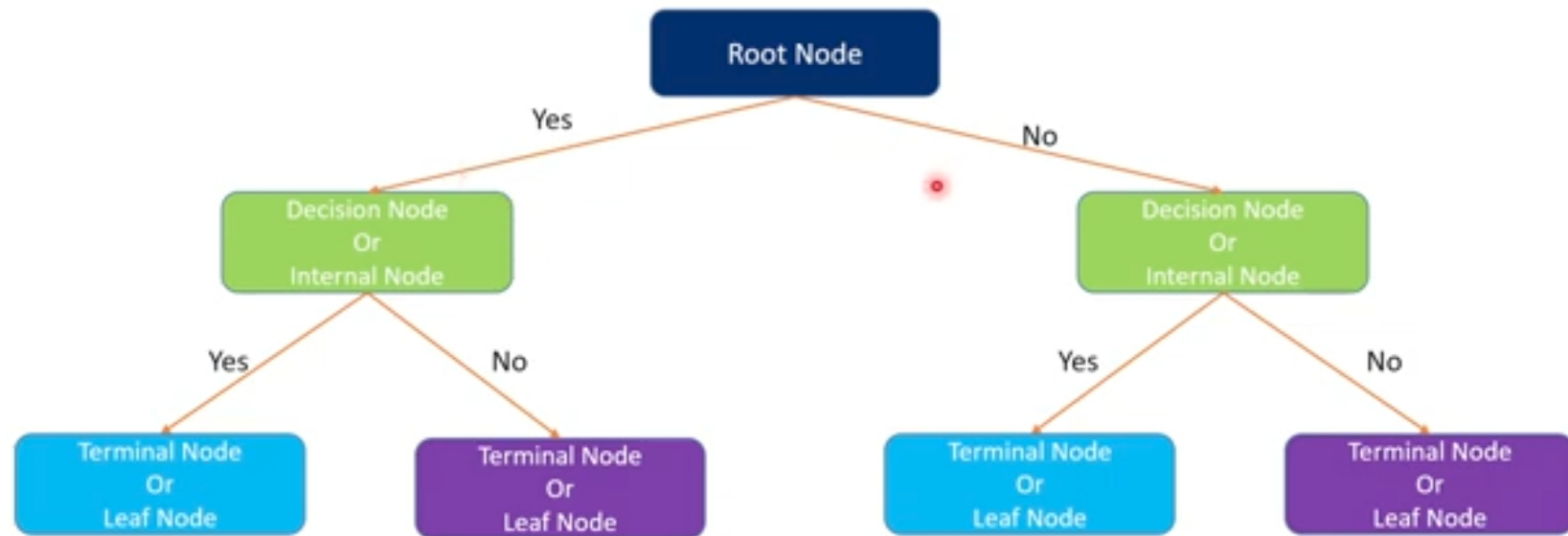


Picture credits: AI Time Journal



Picture credits:
<https://towardsdatascience.com/decision-tree-hugging-b8851f853486>

Decision Tree - Terminologies



Decision Tree

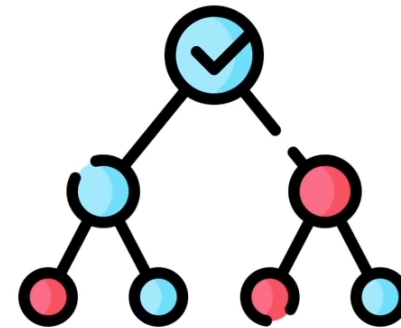
Advantages:

1. Can be used for both Classification & Regression
2. Easy to interpret
3. No need for normalization or scaling
4. Not sensitive to outliers

Disadvantages:

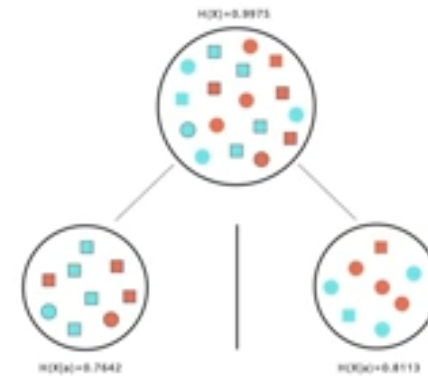
1. Overfitting issue
2. Small changes in the data alter the tree structure causing instability
3. Training time is relatively higher

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Entropy, Information Gain & Gini Impurity



Decision Tree

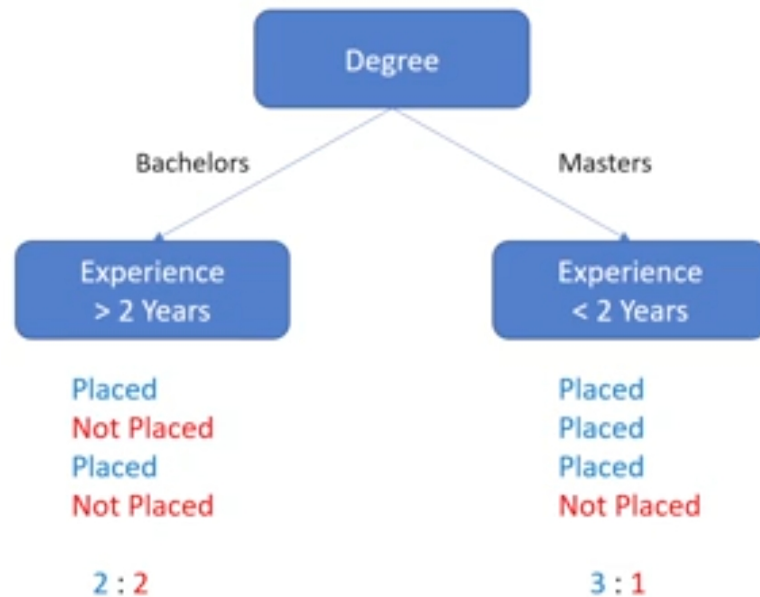
Problem Statement: Build a Decision Tree to determine whether a person will **get a Job** or not based on their **Degree & Years of Experience**.

Degree	Experience in Years	Placed / Not Placed
Masters	2	Placed
Bachelors	0	Not Placed
Masters	3	Placed
Masters	1	Not Placed
Bachelors	2	Placed
Masters	3	Placed
Bachelors	0	Not Placed
Bachelors	1	Not Placed

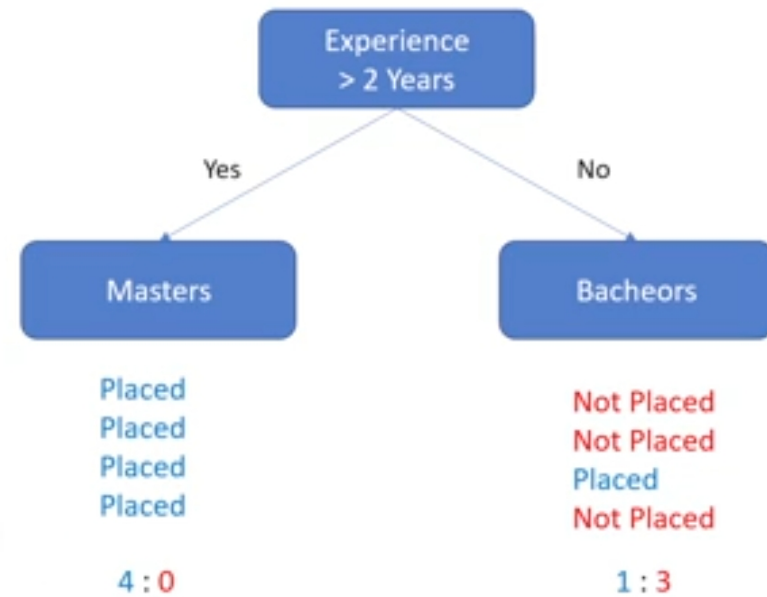
Decision Tree



Decision Tree



Entropy: High
Information Gain: Low
Gini Impurity: High



Entropy: Low
Information Gain: High
Gini Impurity: Low

Entropy

Entropy:

In Machine Learning, **Entropy** is the quantitative measure of the **randomness** of the information being processed.

A **high value of Entropy** means that the **randomness** in the system is **high** and thus making accurate predictions is tough.

A **low value of Entropy** means that the **randomness** in the system is **low** and thus making accurate predictions is easier.



Low Entropy



High Entropy

$$\text{Entropy} = \sum_{i=1}^c -p_i \log_2 p_i$$

c --> number of classes

p_i --> Probability of i^{th} class

Information Gain

Information Gain is the measure of how much information a feature provides about a class. Low entropy leads to increased Information Gain and high entropy leads to low Information Gain.

Information gain computes the difference between **entropy before split** and average entropy **after split** of the dataset based on a given feature.

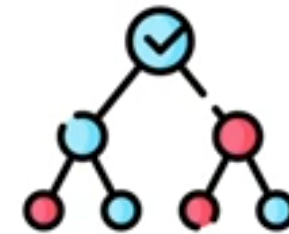


$$\text{Information gain (T, F)} = \text{Entropy (T)} - \sum_{v \in F} \frac{|T_v|}{T} \cdot \text{Entropy (T)}$$

Gini Impurity

The split made in a Decision Tree is said to be pure if all the data points are accurately separated into different classes.

Gini Impurity measures the likelihood that a randomly selected data point would be incorrectly classified by a specific node.



$$G = \sum_{i=1}^C p(i) * (1 - p(i))$$

